

Shivanshu Maurya

Kanpur, Uttar Pradesh, India | +91-6306899902
shivanshu2211maurya@gmail.com
linkedin.com/in/shivanshumaurya2203 | github.com/Shivanshu-2205
leetcode.com/u/shivanshu2211maurya | shivanshu-portfolio-swart.vercel.app

Summary

Aspiring software engineer seeking to apply problem-solving and backend development skills to build scalable, reliable software solutions while continuously learning and contributing to impactful projects.

Technical Skills

- **Languages:** Python, C++, JavaScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems
- **Software Engineering:** Node.js, Express.js, FastAPI, REST APIs, React, Socket.io
- **Testing & Automation:** Jest, Postman, Unit/Integration Testing, GitHub Actions (CI/CD)
- **Tools & Platforms:** VS Code, Git, Docker
- **Databases:** PostgreSQL, MongoDB, Redis

Projects

- NoteWeave AI | [GitHub](#) | [Live](#)

Jan 2026 – May 2026

Node.js, Express.js, MongoDB, Redis, BullMQ, Gemini API

 - Designed and built scalable backend REST APIs with Node.js and Express.js, structuring services around MongoDB for persistence and Redis/BullMQ for asynchronous job processing.
 - Implemented retry mechanisms and background job queues with BullMQ and Redis to reliably process long-running document and embedding workloads.
 - Built a document Q&A and knowledge-graph feature using RAG with in-memory cosine similarity search, avoiding a dedicated vector database while keeping query latency low.
 - Wrote unit and integration tests for API endpoints using Jest, covering payment and job-queue edge cases to improve code reliability.
- DriftGuard (Observability & Self-Healing Platform) | [GitHub](#)

May 2026 – Present

Node.js, Express.js, PostgreSQL, Prisma, Async Services

 - Developed a backend observability platform using Node.js, Express.js, PostgreSQL, and Prisma to monitor production ML workloads.
 - Built asynchronous services for automated anomaly detection, alerting, and self-healing workflows, focused on improving system reliability and uptime.
 - Implemented statistical drift-detection methods including PSI, KL Divergence, KS Test, Chi-Square, Page-Hinkley, Z-score, IQR, and EWMA to detect anomalies across monitored data streams.
 - Added unit tests for drift-detection statistical functions to verify calculation correctness and edge-case handling.
- Rabbit Hole Explorer (Backend Service for Content Discovery) | [GitHub](#) | [Live](#)

Nov 2025 – Jan 2026

Node.js, Redis, MongoDB, React

 - Built backend services in Node.js for graph traversal, ranking, and recommendation generation over a MongoDB-backed knowledge graph.
 - Designed the service layer for scalability under concurrent workloads, using Redis caching to reduce response latency.
 - Handled API/service design for content discovery and user interaction tracking across a growing dataset.

Education

Pranveer Singh Institute of Technology	2023 – 2027
B.Tech in Computer Science and Engineering — Percentage: 81.02%	
B.L.M. Inter College (U.P. Board)	2021 – 2022
Class XII — Percentage: 87%	
B.L.M. Inter College (U.P. Board)	2019 – 2020
Class X — Percentage: 84.83%	

Achievements and Certifications

- Solved 1000+ Data Structures & Algorithms problems on LeetCode and HackerRank
- 6-star in problem solving using C++ on HackerRank
- ICPC 2023 participant — solved problems under strict time and team constraints
- Smart India Hackathon 2025 participant — built a full-stack solution in a 36-hour hackathon
- Machine Learning Specialization — DeepLearning.AI